MIE 286 Winter 2026: Term Project

Topic: The Effect of Feedback Type on Task Performance

Key Logistics Summary

- **Team Size:** Groups of three (3) students.
- **Team Formation Due:** February 4th, 11:59 PM.
- **Proposal Due:** February 15th, 11:59 PM (20%).
- **Final Report Due:** April 7th, 11:59 PM (80%).
- **Software:** Analysis must be conducted using **R/R Studio**.

PROJECT OVERVIEW

This project is designed to deepen your understanding of course material, experimental design, and research methods. You will pursue a research question of your choice, collect data, and answer it using statistical evidence.

1 PROJECT TEAM

The project is to be completed in **groups of three (3)**.

- **Finding a Group:** If you are having difficulty forming a group, use the course's discussion board on Piazza to find other students. If you are having difficulties, please talk to your project TAs.
- **Conflict Resolution:** If you experience any team issues, contact the project TAs and instructor **as soon as possible**. Do not wait until the report submission deadline to raise these issues.

2 EXPERIMENT DESIGN REFRESHER

Experiments are a key part of scientific and engineering knowledge because they allow for hypothesis testing and the establishment of causal relationships. In planned experiments, we test the relationships between variables by manipulating one or more factors (the **Independent Variables**) while keeping all other factors constant (controlled variables). We then observe changes in one or more outcome variables (the **Dependent Variables**). If all other factors are well-controlled, we can attribute changes in the dependent variable(s) directly to the manipulation of the independent variable(s).

3 PROJECT DESCRIPTION

In engineering systems and User Interface (UI) design, operators are often presented with real-time data about their actions/performance. For example, a factory worker might see a live count of "parts produced per hour," or a gamer might see a ticking countdown timer.

While feedback is generally intended to motivate, inform, or guide the user, the **specific type or modality of feedback** used can significantly alter performance. Different forms of feedback place different demands on the user. For instance, visual feedback might compete with visual tasks, while auditory feedback might be less intrusive but harder to process precisely. This phenomenon is related to *Cognitive Load Theory*—processing different feedback types requires different mental resources.

Your Task: Design an experiment that investigates the **effect of Feedback Type** on the speed and accuracy of a task of your choice. To accomplish this, you must make the following design choices:

3.1 Context

Choose an area you are interested in exploring. The context should be interesting, **not risky** for your participants, and address a research question without an obvious answer. You must generate a specific hypothesis about the relationship between your independent and dependent variables.

- *Example Research Question:* Does **auditory feedback** (e.g., a beep) result in faster typing speed than **visual feedback** (e.g., a visual counter)?

Note: Try to identify a problem or question grounded in the real world.

3.2 Dependent Variables (Outcome Variables)

You must measure two dependent variables:

1. **Response Time (Speed):** E.g., The time taken to complete a puzzle or type a paragraph.
2. **Accuracy (Quality):** E.g., The number of errors made, or a score achieved.

There should be enough granularity in your measures of speed and accuracy so that they can be treated as **continuous** variables in your analysis.

3.3 Independent Variable (The Factor)

Your independent variable will be a single categorical factor called **"Feedback Type"** with two distinct levels. You must compare two *active* forms of feedback. **Do not** use a "No Feedback" condition. You will have to make decisions regarding the differing characteristics, such as modality (Visual vs. Auditory), timing (Immediate vs. Delayed), or content (Detailed vs. Simple).

- **Level 1. Feedback Type A:** The participant performs the task receiving feedback in a specific form (e.g., Visual Feedback).

- **Level 2. Feedback Type B:** The participant performs the task receiving feedback in a different form (e.g., Auditory Feedback).

3.4 Experimental Method

You must design an experiment removing systematic biases from your design as much as possible. This includes:

- How you manipulate the independent variable (how you present the different feedback types).
- How you assign conditions to participants (e.g., random assignment).
- How you control for other factors (e.g., ensuring all participants receive identical instructions).

3.5 Participants

Your participants should be representative of the study population you wish to examine.

- **Sample Size:** You must collect data from at least **12 different participants** for *each* level of your independent variable (Total $N \geq 24$).
- **Sampling Method:** You may collect data from family and friends, but you must discuss the limitations associated with convenience sampling in your report.

4 ANALYSIS METHODS

Describe your data using descriptive statistics and draw conclusions using appropriate inferential statistics (e.g., t-tests, correlation). Analyze the relation between speed and accuracy.

> The **speed-accuracy trade-off** describes the relationship between how fast someone might respond and how accurate they might be.[1]
>
> **Requirement:** You must use **R/R Studio** (and no other software) to conduct your analyses.

5 DISCUSSION AND CONCLUSIONS

Discuss what your results mean in terms of your overall research context, and regarding the trade-off between your two dependent variables (speed and accuracy).

You should also describe any limitations to your study that arise because of the way your experiment was designed or how your data was collected (e.g., your results might have varied due to having different experimenters operating the stopwatch). Also discuss lessons learned and how you may have run the experiment differently if you were to do it again with the same resources and if you had more resources.

[1] See Heitz, R. P. (2014). The speed-accuracy tradeoff: history, physiology, methodology, and behavior. *Frontiers in Neuroscience*. Available at: http://www.ncbi.nlm.nih.gov/pmc/articles/PMC4052662/

6 DELIVERABLES

There are two major deliverables. For each day a deliverable is late, the assessed mark is reduced by 20%.

6.1 1. Project Proposal (4% of course grade)

Due: February 15th at 11:59 PM (Submit via Quercus)

Your proposal should describe what you plan to do, specifically:

- The specific hypotheses you want to investigate.
- Your experimental design and data collection methods.
- A description of the variables to be measured and a brief analysis plan.

Format: A **two-page document**, at least 1.5 line spacing, 1-inch margins, and 12-point font. All required elements should be included within the page limit.

6.2 2. Final Report (15% of course grade)

Due: April 7th at 11:59 PM (Submit via Quercus)

Submit only one report per group. It must be written in the format of a formal research report containing:

1. **Introduction:** Goal of the study, research questions, and background information.
2. **Methods:** Experimental procedure, description of participants/apparatus, and variable definitions. Discuss possible biases.
3. **Analyses:** Description of the statistical tests performed.
4. **Results:** Descriptive and inferential statistics, including assumption checks (e.g., normality). Use tables/figures effectively.
5. **Conclusion:** Conclusions based on statistical analysis, recommendations for further study.
6. **References:** Include any papers or works you need to cite here. As to the format of the citations, any style is acceptable as long as it is used consistently.
7. **Appendices:** Detailed R code and raw statistical outputs (e.g., normality checks).

Format: Maximum **10 pages** (excluding Appendix, References, and Title Page). At least 1.5 line spacing, 12-point font. An attribution table form must be included.